Acknowledgement:

(c) Partially resolved or unresolved, but the remaining concerns are not easily addressed in a short rebuttal - Please select this option sparingly and only when you believe that your questions concern the core tenets of the work, and addressing them requires a significant update to the paper.

Reasons:

Thank you for the rebuttal. It provides additional clarification and further arguments in response to the comments. However, the main concerns remain unresolved, including the necessity of the complex pipeline design, the risk of error propagation, and the limited scope of the evaluation.
In addition, I believe the paper would require substantial structural revision to improve clarity and completeness. For example, the absence of a related work section in the main text makes it harder to position the contribution appropriately. Overall, fully resolving these concerns would require substantial additional work on the paper.

Reply Rebuttal Comment for Reviewer pde4:

(1) Cardiac function assessment is inherently a multimodal, multiscale task requiring four-chamber synergistic mechanisms (ventricular interdependence, atrioventricular synchrony, right-left heart coupling). We designed four encoding streams: semantic (text alignment), temporal (phase context), visual (macro spatial patterns), and dynamic (micro kinematic details). Ablation (Table 2) quantifies independent contributions: removing semantic stream increases RMSE to 6.96 (Δ=2.83), visual to 4.49 (Δ=0.36), dynamic to 4.25 (Δ=0.12), coarse-grained layer to 7.21 (Δ=3.08), fine-grained layer to 4.68 (Δ=0.55). Thus each component provides non-redundant gains, with four-chamber synergy contributing most. State-of-the-art works (EchoPrime, PanEcho, Echo-Vision-FM) also adopt multi-view or multi-task fusion, aligning with our design. Moreover, the full model achieves narrower Bland-Altman limits and better edge-case performance, demonstrating robustness under modality degradation. Hence, our framework's complexity is deliberate, not arbitrary.

(2) We acknowledge validation on a single dataset is limited. EchoNet-Dynamic (10,030 videos, 8:1:1 patient-wise split) is the largest public EF benchmark, enabling fair comparison with prior methods and validating our core innovations. However, real-world clinical diversity (equipment, operators, views, tasks) requires broader evaluation. In the revised manuscript, we will: (i) collect ≥500 multi-device clinical videos from two tertiary hospitals; (ii) evaluate on EchoNet-LVH (PLAX) and CAMUS (A2C, A4C); (iii) release code and pre-trained weights. We will add a "Generalization and Future Work" subsection and conclude that cross-dataset, multi-center validation is core future work.

(3) To quantify error propagation, we reduced the left ventricular segmentation mask to 9/10, 7/8, and 5/6 of its original area. Even a 10% reduction (9/10) causes catastrophic degradation: MAE rises from 2.95 to 9.31, RMSE from 4.13 to 12.31, R^2 from 0.88 to −0.0012. At 7/8: MAE=19.69, RMSE=12.41, R^2=−1.99. At 5/6: MAE=20.32, RMSE=22.35 R^2=−2.18. Thus segmentation errors severely propagate and worsen EF prediction. To address this, we plan: (i) a

segmentation quality monitoring module to detect mask anomalies; (ii) end-to-end joint training to co-optimize segmentation and prediction; (iii) segmentation mask perturbations as data augmentation. These plans will be detailed in the Conclusion and Future Work section.

(4) Thank you for your comment. To address the lack of a dedicated related work section, we have integrated a comprehensive literature review directly into the Introduction (Section 1). This review systematically examines existing echocardiographic EF prediction methods from three perspectives: fusion mechanisms, semantic alignment, and physiological modeling, and clearly identifies their limitations. Consequently, a separate "Related Work" chapter is no longer necessary. Detailed bibliographic notes and extended discussions have been moved to the Appendix, and the complete reference list, along with supplementary materials, can be accessed via the repository link provided in the manuscript (https://github.com/EF-VLNET-2026/EF-VLNet). This structural revision improves the conciseness and completeness of the main text while preserving full transparency. We appreciate your valuable suggestion.

"1. Introduction

Cardiovascular disease remains a leading cause of mortality worldwide (Zhao et al., 2019), and accurate assessment of cardiac function is essential for diagnosis, treatment planning, and prognosis. Left ventricular ejection fraction (LVEF) is the most widely used clinical metric for evaluating systolic function. Although echocardiography is the preferred imaging modality due to its portability and real-time capability (Kirkpatrick et al., 2020), automated LVEF prediction from ultrasound videos remains challenging because of low image quality, speckle noise, and inter-operator variability.

A major source of these challenges lies in the fact that existing echocardiographic analysis methods predominantly focus exclusively on the left ventricle, neglecting the synergistic mechanisms among all four cardiac chambers that are essential for accurate cardiac function assessment. As noted by Wever Pinzon et al. (2012) and Naeije & Badagliacca (2017), optimal cardiac function relies critically on mechanical coupling between ventricles, atrioventricular synchrony, and comprehensive interventricular hemodynamic regulation circuits. This oversimplified modeling paradigm often yields predictions that lack physiological plausibility. Furthermore, contemporary feature extraction and fusion methods primarily operate at the data level, failing to establish a coherent mapping system from pixel-level features to clinical semantics, consequently limiting model interpretability and clinical utility. To contextualize the limitations of prior work and to motivate the proposed framework, we systematically review existing echocardiographic EF prediction methods from three perspectives: fusion mechanism, semantic alignment, and physiological modeling.

Early single modality approaches (EchoNet Dynamic, Ouyang et al., 2020; EchoCoTr, Muhtaseb & Yaqub, 2022; EFNet, Batool et al., 2024) predict EF solely from B mode videos and are confined to the left ventricle. Recent multi-modal methods incorporate textual information: Echo Vision FM (Christensen et al., 2024) employs self-supervised contrastive learning,

EchoPrime (Vukadinovic et al., 2024) fuses multi-view videos via anatomical attention, and PanEcho (Holste et al., 2025) outputs 39 multi-task heads. However, these methods primarily rely on global contrastive or retrieval-based aggregation without explicitly distinguishing multi-scale physiological features, and MMnet (Cai et al., 2024) still operates at the data-level fusion stage.

Existing vision-language models (e.g., Echo Vision FM, Christensen et al., 2024; EchoPrime, Vukadinovic et al., 2024) achieve video report alignment via global contrast, making it difficult to establish fine-grained correspondences between specific clinical terms, such as "mild systolic dysfunction," and particular spatiotemporal regions. Chen et al. (2025) combine gaze with VLMs, and MedAligner (Yan et al.) uses word-region alignment, but neither optimizes for the temporal characteristics of cardiac ultrasound. While BioBERT (Lee et al., 2020) and Lingshu 7B (Xu et al., 2025) provide powerful medical language representations, they alone cannot achieve cross-modal fine-grained alignment, and Zamzmi et al. (2020) note that current AI models struggle with specialized terminology such as "segmental wall motion abnormalities," highlighting a significant semantic gap.

Cardiac function depends on synergistic mechanisms, including ventricular interdependence (Wever Pinzon et al., 2012), atrioventricular synchrony (Fortuin et al., 1972), and right-left heart coupling (Naeije & Badagliacca, 2017). Nevertheless, existing VLMs (EchoPrime, Vukadinovic et al., 2024; PanEcho, Holste et al., 2025) as well as traditional methods (EchoNet Dynamic, Ouyang et al., 2020; EchoCoTr, Muhtaseb & Yaqub, 2022; EFNet, Batool et al., 2024) focus predominantly on the left ventricle or single-view analysis. MedSAM2 (Ma et al., 2025) achieves strong segmentation performance but does not build a multimodal fusion architecture; Kang et al. (2022) and Blaivas & Blaivas (2022) similarly do not incorporate four-chamber synergy.

In summary, current work has not systematically encoded inter-chamber synergistic mechanisms into model designs, predominantly adopts global contrastive fusion without explicit multi-scale differentiation, and limits semantic alignment to global video report matching. To address these critical challenges, this paper presents a novel framework for cardiac function assessment that leverages vision language models and multi-scale semantic fusion. Our fundamental innovation involves modeling the heart as an integrated, multi-scale semantic system, explicitly constructing cross-hierarchical feature representations that span from coarse-grained chamber dynamics to fine-grained tissue motion using vision-language models. This approach enables precise alignment of visual features with professional medical language descriptions within a unified semantic space. The main contributions of this work are as follows:

1. *We propose a novel multi-scale semantic fusion framework that employs hierarchical feature learning mechanisms to simultaneously capture macroscopic dynamic characteristics and microscopic motion patterns of the heart, establishing sophisticated semantic relationships between coarse-grained and fine-grained features through attention-based fusion modules.*

2. *We present the first systematic integration of vision language models into cardiac ultrasound analysis, achieving deep semantic alignment between visual features and clinical terminology through medically guided feature embedding, significantly enhancing model interpretability and clinical relevance.*

3. *We design an end-to-end trainable architecture that maintains physiological plausibility*

while achieving precise mapping from raw ultrasound videos to EF prediction. Extensive experiments on public datasets demonstrate superior performance compared to state of the art methods, particularly in handling complex clinical cases.”

Supplementary References:

- *lste G, Oikonomou E K, Mortazavi B J, Wang Z, Khera R. Complete AI-enabled echocardiography interpretation with multitask deep learning. JAMA, 2025, 334(4): 306-318.*
- *Blaivas M, Blaivas L. Machine learning algorithm using publicly available echo database for simplified "visual estimation" of left ventricular ejection fraction. World Journal of Experimental Medicine, 2022, 12(2): 16.*